Sharp uniform Landis estimates for the biharmonic equation

Abstract. We determine the sharp uniform Landis scale for the biharmonic equation with bounded potentials. For every sufficiently large R and every $n \geq 2$, we construct smooth real-valued functions u_R, V_R satisfying $\Delta^2 u_R + V_R u_R = 0$, $\|V_R\|_\infty \leq 1$ and $u_R(0) = \|u_R\|_\infty = 1$, such that $\|u_R\|_{L^\infty(B_1(x_R))} \leq \exp(-cR^{4/3}\log R)$ at some point $|x_R| = R$. Together with the lower bound of Huang, Wang and Zheng, this establishes sharpness of the full scale $R^{4/3}\log R$ in both real and complex scalar classes. In the plane, the potentials are compactly supported. The construction uses an exact biharmonic seed, adaptive removal of its poles, and a fourth-order correction along a transverse real-part zero set. Kelvin inversion turns the resulting annular connection into a high-order zero; dilation produces the logarithmic factor and places the observation ball at the prescribed radius. The examples depend on R: the conclusion concerns a uniform bound over normalized pairs, not all-direction decay of a single fixed solution.

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1. Introduction

Quantitative unique continuation at infinity asks how small a normalized solution of an elliptic equation can be far from the normalization point. For bounded potentials, the answer depends on both the operator and the scalar class. In this paper we determine the sharp uniform scale for the biharmonic equation

$$\Delta^2 u + Vu = 0 \quad \text{in } \mathbb{R}^n, \qquad n \geq 2, \qquad \Delta = \sum_{j=1}^{n} \partial_j^2. \tag{1.1}$$

For a bounded solution with $u(0) = 1$, set

$$m_u(R) = \inf_{|x_0|=R} \|u\|_{L^\infty(B_1(x_0))}. \tag{1.2}$$

The infimum records the smallest size of the solution on a unit ball at distance R from the origin. Huang, Wang and Zheng [14, Theorem 1.1] proved that, if $\|V\|_\infty \leq 1$ and $\|u\|_\infty \leq C_0$, then

$$m_u(R) \geq \exp(-CR^{4/3}\log R), \qquad R \geq R_0, \tag{1.3}$$

where C, R_0 depend only on n, C_0. Their theorem permits complex-valued solutions and potentials and applies to all positive integer powers of the Laplacian. A fixed multiplicative constant in their estimate has been absorbed into C, R_0.

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We show that the full scale in (1.3) is optimal, including the logarithmic factor, even when both the solution and the potential are real. The construction attains this scale at every prescribed sufficiently large radius. Since the resulting pairs belong to the real class, they establish sharpness in the larger complex classes as well. Thus real-valuedness is a strength of the examples, not a restriction on the scope of the sharpness conclusion.

1.1. **Main results**

Theorem 1.1. *There are constants $c > 0$ and $R_* > 1$ such that, for every $n \geq 2$ and every real number $R \geq R_*$, there exist $u_R, V_R \in C^\infty(\mathbb{R}^n;\mathbb{R})$ and $x_R \in \mathbb{R}^n$ satisfying*

$$\Delta^2 u_R + V_R u_R = 0, \qquad \|V_R\|_\infty \leq 1, \qquad u_R(0) = \|u_R\|_\infty = 1, \tag{1.4}$$

and

$$|x_R| = R, \qquad \|u_R\|_{L^\infty(B_1(x_R))} \leq \exp(-cR^{4/3}\log R). \tag{1.5}$$

When $n = 2$, the potential V_R is compactly supported, and u_R is a nonzero constant outside a sufficiently large ball.

To express sharpness uniformly over the real and complex classes, let

$$\mathfrak{T} = \{(\mathbb{R},\mathbb{R}), (\mathbb{C},\mathbb{R}), (\mathbb{C},\mathbb{C})\}.$$

The first field specifies the values of the solution, and the second those of the potential. For $(\mathbb{F},\mathbb{K}) \in \mathfrak{T}$ and $C_0 \geq 1$, let $C_n^{\mathbb{F},\mathbb{K}}(C_0)$ consist of the pairs

$$\begin{aligned} &u \in L^\infty(\mathbb{R}^n;\mathbb{F}), \qquad V \in L^\infty(\mathbb{R}^n;\mathbb{K}), \qquad \|V\|_\infty \leq 1, \\ &\Delta^2 u + Vu = 0 \text{ in distributions}, \qquad u(0) = 1, \qquad \|u\|_\infty \leq C_0. \end{aligned} \tag{1.6}$$

The distributional formulation does not introduce an ambiguity in the normalization. Interior elliptic regularity gives $u \in W^{4,p}_{\mathrm{loc}}$ for every $1 < p < \infty$; in particular, for any $x \in \mathbb{R}^n$,

$$\|u\|_{W^{4,p}(B_1(x))} \leq C(n,p)\big(\|u\|_{L^p(B_2(x))} + \|Vu\|_{L^p(B_2(x))}\big).$$

Taking $p > n$ gives the continuous representative used at the origin. The same regularity justifies smooth approximation of compactly supported cutoff products in $W^{4,2}$ in the estimates underlying (1.3); no derivative of V is needed. The corresponding extremal exponent is

$$\mathcal{E}_n^{\mathbb{F},\mathbb{K}}(R;C_0) = \sup_{(u,V)\in C_n^{\mathbb{F},\mathbb{K}}(C_0)} [-\log m_u(R)]. \tag{1.7}$$

Corollary 1.2. *For every $n \geq 2$ and $C_0 \geq 1$, there are constants $C, R_0 > 0$, depending only on n, C_0, such that, simultaneously for all $(\mathbb{F},\mathbb{K}) \in \mathfrak{T}$,*

$$cR^{4/3}\log R \leq \mathcal{E}_n^{\mathbb{F},\mathbb{K}}(R;C_0) \leq CR^{4/3}\log R \qquad (R \geq R_0). \tag{1.8}$$

Consequently, if $\Psi : [2,\infty) \to [0,\infty)$ satisfies $\Psi(R) = o(R^{4/3}\log R)$, no lower bound $m_u(R) \geq \exp(-C\Psi(R))$ can hold for all sufficiently large R uniformly over any of these classes.

Proof. The upper bound follows from (1.3). For the lower bound, the examples in Theorem 1.1 lie in $C_n^{\mathbb{R},\mathbb{R}}(1)$ and hence in every class under consideration. A proposed uniform estimate with Ψ, applied to the example at the same radius R, would give $cR^{4/3}\log R \leq C\Psi(R)$, contrary to the assumed little-oh condition. □

Remark 1.3. The order of quantifiers is $\forall R \geq R_*\ \exists(u_R, V_R)$, not $\exists(u,V)\ \forall R \geq R_*$. In particular, the result does not provide a fixed pair with this smallness along an unbounded sequence, nor a solution with this decay in every direction. Each planar potential is compactly supported, but the supports need not lie in a common fixed compact set. No sign condition or uniform derivative bound is imposed on the potentials.

1.2. **Background and related results**

The result sits at the intersection of three questions: local vanishing, decay of a fixed solution at infinity, and lower bounds uniform over a normalized family. These questions are closely related by rescaling, but their normalizations and quantifiers are different. We review the distinction before explaining the construction.

Local unique continuation and quantitative bounds. For second-order equations, the qualitative theory of unique continuation for Schrödinger operators was developed in particular by Jerison and Kenig [15]. A geometric approach through frequency and doubling estimates was established by Garofalo and Lin [12, 13]. In the spectral setting, the work of Donnelly and Fefferman [8] relates the order of vanishing of an eigenfunction to its spectral parameter. These results provide two complementary points of view: weighted estimates propagate nontriviality, while frequency measures the rate at which a solution can vanish.

For equations with a variable potential, quantitative estimates must also specify how constants depend on that potential and on the normalization of the solution. This dependence is central to the work of Kukavica [18], the account of Kenig [16], and the Carleman estimates of Bakri [2]. It is essential to keep the normalization visible: harmonic polynomials of arbitrarily high degree already show that the potential norm alone cannot bound the vanishing order of an unrestricted local solution. Bounds for regular potentials, such as those of Bakri [1] and Zhu [25], measure additional coefficient regularity. They therefore concern a different uniformity question from one in which only $\|V\|_\infty$ is controlled.

From Landis' question to a unit-ball estimate. The classical Landis question concerns whether a nonzero solution of a Schrödinger equation with bounded potential can decay faster than an exponential at infinity. In the complex-valued class, Meshkov [24] constructed a bounded potential and a nontrivial solution in the plane with decay of order $\exp(-c|x|^{4/3})$. His work identifies the power $4/3$ as the critical one for that complex-valued decay problem. Motivated by Anderson localization, Bourgain and Kenig [3] obtained the quantitative unit-ball lower bound with exponent $CR^{4/3}\log R$ for bounded normalized solutions of the second-order equation. Extensions of unique continuation at infinity to other coefficient regimes include Cruz-Sampedro [5] for complex potentials, Davey [6] for magnetic Schrödinger operators, and Lin and Wang [21] for general second-order elliptic equations.

The logarithm distinguishes the unit-ball problem from the critical-power question. Under a large-scale rescaling, a fixed observation ball becomes a small ball; a power-law lower bound on that ball then produces a logarithmic factor. Conversely, an example bounded above by $\exp(-cR^{4/3})$ does not exclude a uniform lower bound $\exp(-CR^{4/3})$ with a larger constant C. Establishing that the logarithm is necessary requires a smaller example, with the global normalization retained.

The real-valued problem and the role of dimension. In the plane, real-valuedness leads to a different second-order theory. Kenig, Silvestre and Wang [17] proved a quantitative Landis estimate under a nonnegative Schrödinger potential; the sign convention here is $-\Delta w + Ww = 0$ with $W \geq 0$. Davey, Kenig and Wang [7] extended this line of analysis to variable-coefficient planar equations. The sign-free real problem was subsequently treated by Logunov, Malinnikova, Nadirashvili and Nazarov [23]. For the comparison relevant here, their Theorem 1.2 implies that a real entire solution of $\Delta w + Ww = 0$ in $\mathbb{R}^2$, with W real, $\|W\|_\infty \leq 1$ and $w(0) = \|w\|_\infty = 1$, satisfies

$$\inf_{|x_0|=R} \|w\|_{L^\infty(B_1(x_0))} \geq \exp\bigl(-CR(\log R)^{3/2}\bigr)$$

for all sufficiently large R. This follows by applying their local theorem on B_{4R} with its radius parameter equal to $2R$. Their theorem on decay in every direction is a separate statement; it should not be identified with this unit-ball estimate.

The dimension restriction is substantive. In a recent preprint, Frank and Ivanisvili [10, Theorem 1] construct, in every dimension $n \geq 3$, a nonzero smooth real solution with a smooth bounded real potential and all-direction decay $\exp(-c|x|^{4/3})$. Thus a general contrast between real second-order and fourth-order equations would be inaccurate. The contrast established here is already present in the plane: real-valuedness improves the second-order unit-ball bound beyond the power $4/3$, whereas for the

biharmonic operator the full scale $R^{4/3}\log R$ remains necessary. Our construction then extends to all dimensions $n \geq 2$.

A related, but distinct, local result is due to Feng, Gan, Jiang and Lin [9, Theorem 1.1]. For every sufficiently large integer k, they construct real Dirichlet solutions in $B_3 \subset \mathbb{R}^3$ with vanishing order k and potential norm at most $Ck^{3/2}$. This is a fixed-domain second-order construction. It does not directly supply the bounded entire solutions, the maximum normalization, or the prescribed distant observation ball required here. Neither this result nor the second-order examples above are used as inputs to our proof.

Powers of the Laplacian. For higher-order equations, unique continuation must be studied with the structure of the principal part in view. Strong unique continuation for powers of the Laplacian was investigated by Colombini and Grammatico [4] and Lin [19]. Lin, Nagayasu and Wang [20] developed quantitative estimates with singular lower-order coefficients, including three-sphere and doubling inequalities. Further estimates for vanishing orders, with explicit coefficient dependence, were obtained by Zhu [25, 26]. More recently, Liu, Tian and Yang [22] studied quantitative unique continuation for a bi-Laplacian equation with a $W^{1,\infty}$ potential. These derivative-sensitive estimates do not give a uniform improvement over a class in which no derivative of the potential is bounded.

On the counterexample side, Fu, Zhang and Zuazua [11] constructed complex-valued biharmonic examples with decay of order $\exp(-c|x|^{4/3})$ in connection with the optimality of observability inequalities for plate equations. On the lower-bound side, Huang, Wang and Zheng [14] established (1.3). The present construction matches that lower bound in the smallest scalar class, with smooth real solutions and smooth real potentials. It addresses the logarithmic loss that is not settled by a critical-power decay example. Smoothness here is qualitative: the potentials have a uniform L^∞ bound, but no uniform derivative bound or common compact support is asserted.

1.3. **Outline of the construction**

The argument begins in the plane, where the identity $\Delta^2 = 16\partial^2\bar{\partial}^2$ provides exact biharmonic products. At degrees $k = q^2$, a polynomial with separated roots determines a complex seed whose inner and outer models are $A_k|z|^2$ and $|z|^2 z^{-k}$. Matching on fixed collars introduces only an exponentially small residual. The poles at the roots must then be removed before a real potential can be defined.

Taking a real part is not sufficient: the numerator $\Delta^2 \operatorname{Re} F$ need not vanish on the zero set of $\operatorname{Re} F$. Section 2 resolves this obstruction where that zero set is transverse. With $h = \operatorname{Re} F/|F|$, a correction of order h^4 preserves the first three jets and cancels the fourth-order residual on $h = 0$. A quantitative gradient bound then gives smooth division and a bounded real quotient. In the root cores, by contrast, the replacement remains exactly biharmonic. Possible critical zeros there require no division. The argument uses both mechanisms, not the Taylor correction alone.

Sections 3 and 4 arrange the geometry needed by this correction. The two replacement radii $\varepsilon_0 k^{-1/2}$ and $\varepsilon_1 k^{-2/3}$ distinguish roots where the holomorphic phase dominates from those where the replacement phase must dominate. This avoids cancellation of phase gradients and yields the decisive bound $D_\xi \leq Ck^3$. Disjoint residual supports then give a real annular connection with $\|Q_k\|_\infty \leq Ck^3$, without a loss proportional to the number of roots (Proposition 4.3).

Finally, Section 5 applies Kelvin inversion. The outer model becomes a harmonic polynomial of degree k near zero, while the inner model becomes a nonzero constant near infinity. A dilation of order $k^{3/4}$ normalizes the potential. On the observation ball, the polynomial contributes the factor $\lambda^{-k} = \exp(-k\log\lambda)$. Choosing $k \asymp R^{4/3}$ and $\lambda \asymp R$ gives precisely the scale $R^{4/3}\log R$. Translation from a point of maximum modulus and the final choice of dilation place the center at exactly the prescribed radius, not merely along a sequence.

2. A real-part correction

The correction preserves the real part through third order on its zero set, while its fourth normal jet cancels the biharmonic residual there through the identity $\Delta^2(h^4a)|_{h=0} = 24a|\nabla h|^4$. When phase transversality makes ∇h non-degenerate, this cancellation permits smooth division by the corrected real function, without requiring that function to be nonvanishing. The estimate below quantifies this step; exact biharmonicity will handle the critical zeros outside the correction regions.

2.1. Notation and biharmonic products

We identify $\mathbb{R}^2$ with $\mathbb{C}$ and write $z = x_1 + ix_2$. The notation $|D^jF|$ denotes the maximum of $|\partial^\alpha F|$ over real multiindices α with $|\alpha| = j$; gradient norms are Euclidean. Unless indicated otherwise, constants depend only on fixed radii, cutoff functions and the stated finite derivative orders, never on q, k or an individual root. We write $X \asymp Y$ when $cY \le X \le CY$ with such fixed positive constants.

Set

$$\partial = \frac{1}{2}(\partial_{x_1} - i\partial_{x_2}), \qquad \bar\partial = \frac{1}{2}(\partial_{x_1} + i\partial_{x_2}), \qquad \Delta^2 = 16\partial^2\bar\partial^2.$$

If H is holomorphic and G is smooth, then

$$\Delta^2(HG) = 16\left(H''\bar\partial^2 G + 2H'\partial\bar\partial^2 G + H\partial^2\bar\partial^2 G\right). \tag{2.1}$$

Since $\bar\partial H = 0$, the identity follows by applying $\bar\partial^2$ to G and then using the product rule for ∂^2. In particular, $\bar\partial^2 G = 0$ implies $\Delta^2(HG) = 0$. This exact cancellation will preserve biharmonicity in the replacement cores.

2.2. Correction along a transverse zero set

For a nonvanishing complex-valued smooth function F, set

$$\nabla \arg F := \operatorname{Im}\left(\frac{\nabla F}{F}\right).$$

This definition does not require a single-valued choice of argument on the domain.

Lemma 2.1 (Real-part correction)**.** *Let $\Omega \subset \mathbb{R}^2$ be open and let $F \in C^\infty(\Omega;\mathbb{C})$ be nonvanishing. Put*

$$A = |F|, \qquad f = \operatorname{Re} F, \qquad h = f/A, \qquad E = \Delta^2 f.$$

Let $L \ge 1$ and $D > 0$. Assume that, for fixed positive constants b, c, C_* and $\{C_j\}_{j=0}^9$, the relative derivative bounds and phase estimate

$$|D^jF| \le C_j L^j A \qquad (0 \le j \le 9), \tag{2.2}$$

$$|D^jE| \le C_j D L^j A \qquad (0 \le j \le 5), \tag{2.3}$$

$$|\nabla \arg F| \ge cL \qquad \text{on } \{x \in \Omega : |h(x)| \le 1/2\}, \tag{2.4}$$

hold, and that the residual has the size and support margin

$$D \le C_* L^4, \qquad \operatorname{supp} E \Subset \Omega, \qquad \operatorname{dist}(\operatorname{supp} E, \mathbb{R}^2 \setminus \Omega) \ge b/L. \tag{2.5}$$

Then there exist $U \in C^\infty(\Omega;\mathbb{R})$ and $Q \in C_c^\infty(\Omega;\mathbb{R})$ such that

$$\Delta^2 U + QU = 0, \qquad \|Q\|_\infty \le CD, \qquad U = f \text{ off } \operatorname{supp} E. \tag{2.6}$$

The ratio U/f extends smoothly with values in $[1/2, 3/2]$, and $U - f$ vanishes to fourth order on $\{f = 0\}$. Here C depends only on b, c, C_* and the constants in (2.2)–(2.3).

Proof. We first record the relative derivative bounds

$$|D^jA| \le C_j' L^j A, \qquad |D^j(A^{-1})| \le C_j' L^j A^{-1}, \qquad |D^j h| \le C_j' L^j \quad (0 \le j \le 9). \tag{2.7}$$

These follow inductively from $A^2 = F\bar F$: the highest multiindex derivative of A has coefficient $2A$, and each lower-order product has the required homogeneity. Differentiating $AA^{-1} = 1$ gives the estimate for

A^{-1}, after which the product rule applied to $h = fA^{-1}$ gives the estimate for h. The constants do not require a positive lower bound for A. Writing $F = Ae^{i\phi}$ locally, we have $h = \cos\phi$; therefore (2.4) yields

$$|\nabla h| = \sqrt{1-h^2}\,|\nabla\phi| \geq c_1 L \quad \text{on } |h| \leq 1/2, \qquad c_1 = \sqrt{3}c/2. \tag{2.8}$$

Choice of the correction. Fix $\chi \in C_c^\infty((-2,2);\mathbb{R})$ with $0 \leq \chi \leq 1$ and $\chi = 1$ on $[-1,1]$. Write $|E| \leq C_E DA$ and set $K = C_E C_*/(3c_1^4)$. Choose once and for all

$$0 < \delta \leq \min\left\{\frac{1}{8}, \frac{c_1 b}{4}, (2K)^{-1/3}\right\}. \tag{2.9}$$

Define

$$B = \chi(h/\delta)\frac{h^4 E}{24|\nabla h|^4}, \qquad U = f - B. \tag{2.10}$$

The quotient is evaluated only on $|h| < 2\delta$, and B is set to zero elsewhere. The gradient bound (2.8) and the compact support of χ make this definition smooth, with $\operatorname{supp} B \subset \operatorname{supp} E$. For $h \neq 0$, the relative correction satisfies

$$\left|\frac{B}{f}\right| \leq \frac{C_E D(2\delta)^3}{24c_1^4 L^4} \leq K\delta^3 \leq \frac{1}{2}.$$

The same ratio extends smoothly across $h = 0$. Consequently,

$$U = Ah\Theta, \qquad \Theta = 1 - \chi(h/\delta)\frac{h^3 E}{24A|\nabla h|^4}, \qquad \frac{1}{2} \leq \Theta \leq \frac{3}{2}, \tag{2.11}$$

with the quotient in Θ interpreted by the same cutoff rule. Thus U and f have exactly the same zero set, and

$$\nabla U = A\nabla h \neq 0 \quad \text{on } \{h = 0\}. \tag{2.12}$$

Cancellation on the zero set. For any smooth functions p, a,

$$\Delta^2(pa) = a\Delta^2 p + p\Delta^2 a + 2\Delta p\,\Delta a + 4\nabla a\cdot\nabla\Delta p + 4\nabla p\cdot\nabla\Delta a + 4\sum_{i,j=1}^{2} p_{ij}a_{ij}.$$

For $p = h^4$, all derivatives of p of order at most three vanish on $h = 0$. Moreover,

$$\Delta(h^4) = 12h^2|\nabla h|^2 + 4h^3\Delta h, \qquad \Delta^2(h^4)\big|_{h=0} = 24|\nabla h|^4.$$

Thus every term involving a derivative of a vanishes on the zero set, leaving precisely

$$\Delta^2(h^4 a)\big|_{h=0} = 24a|\nabla h|^4. \tag{2.13}$$

Since $\chi = 1$ near $h = 0$, applying this identity to (2.10) cancels the residual there:

$$N := \Delta^2 U = 0 \quad \text{on } \{h = 0\}. \tag{2.14}$$

Derivative estimates. On $|h| < 2\delta$, put $v = L^{-1}\nabla h$. By (2.7)–(2.8),

$$|v| \geq c_1, \qquad |D^r v| \leq C_r L^r \quad (0 \leq r \leq 5).$$

Differentiating $|\nabla h|^{-4} = L^{-4}|v|^{-4}$ therefore yields

$$|D^r(|\nabla h|^{-4})| \leq C_r L^{r-4} \qquad (0 \leq r \leq 5). \tag{2.15}$$

This step uses derivatives of h only through order six. Likewise,

$$|D^r[\chi(h/\delta)h^4]| \leq C_{r,\delta} L^r.$$

In a product term with $r + s + t = j$, the three factors in (2.10) are bounded by CL^r, $CDL^s A$ and CL^{t-4}. The product rule thus gives

$$|D^j B| \leq CDL^{j-4}A \qquad (0 \leq j \leq 5). \tag{2.16}$$

Since δ is fixed, its inverse powers enter only the constants. Consequently,

$$|N| \le CDA, \qquad |\nabla N| \le CDLA, \qquad \operatorname{supp} N \subset \operatorname{supp} E. \tag{2.17}$$

The last support assertion uses the locality of differentiation.

Division estimate near the zero set. To divide by U, we must recover the factor $|h|$ missing from the first estimate in (2.17). For $x \in \operatorname{supp} E$ with $0 < |h(x)| < \delta$, set $\sigma = \operatorname{sgn}(h(x))$ and follow the normalized gradient flow

$$\dot\gamma(t) = -\sigma \frac{\nabla h(\gamma(t))}{|\nabla h(\gamma(t))|^2}, \qquad \gamma(0) = x. \tag{2.18}$$

As long as the solution remains in $|h| \le \delta$,

$$h(\gamma(t)) = h(x) - \sigma t, \qquad |\dot\gamma(t)| \le (c_1 L)^{-1}.$$

Up to time $T := |h(x)|$, the path has length at most $T/(c_1 L) \le b/(4L)$. The support margin $\operatorname{dist}(x, \mathbb{R}^2 \setminus \Omega) \ge b/L$ therefore keeps it in a compact subset of Ω, where the gradient is bounded away from zero along the flow. The local ODE solution extends to time T and reaches the zero set: $h(\gamma(T)) = 0$.

The relative amplitude bound in (2.7) gives

$$\left|\frac{d}{dt} \log A(\gamma(t))\right| \le \frac{C_A}{c_1}, \qquad e^{-C_A\delta/c_1} A(x) \le A(\gamma(t)) \le e^{C_A\delta/c_1} A(x).$$

Using (2.14) at the endpoint and integrating (2.17), we obtain

$$|N(x)| \le \int_0^T |\nabla N(\gamma(t))|\, |\dot\gamma(t)|\, dt \le CDA(x)|h(x)|. \tag{2.19}$$

At $h(x) = 0$, the same estimate follows from the cancellation; off $\operatorname{supp} E$, it follows from $N = 0$. On $|h| \ge \delta$, the comparison (2.11) and the residual estimate (2.17) give the quotient bound without a flow argument. Combining the two regions, we obtain

$$|N| \le CD|U| \quad \text{on } \Omega. \tag{2.20}$$

Smooth extension of the quotient. Define $Q = -N/U$ where $U \ne 0$. At a zero, use smooth local coordinates (s, t) with $s = h$, which exist by (2.8). In these coordinates,

$$N(s,t) = s \int_0^1 \partial_s N(\tau s, t)\, d\tau =: s\widetilde{N}(s,t), \qquad U(s,t) = sA(s,t)\Theta(s,t).$$

Hence $Q = -\widetilde{N}/(A\Theta)$ is smooth. The local extensions agree on overlaps because they agree away from $h = 0$, a dense subset of each chart. By continuity, (2.20) gives $\|Q\|_\infty \le CD$ also on the zero set. Moreover, Q vanishes off $\operatorname{supp} E$ and is therefore compactly supported in Ω. The construction is real-valued throughout, and (2.10) gives the claimed fourth-order vanishing. □

Remark 2.2. Smooth division requires transversality of U, or equivalently of h, not of $N = \Delta^2 U$; the numerator need only vanish on the hypersurface $h = 0$. For the quantitative estimate, the phase bound must hold throughout a neighborhood of the residual support with the margin in (2.5). A bound restricted to points where $E \ne 0$ would not justify the flow argument. No such condition is needed at critical zeros lying in open regions of exact biharmonicity.

3. The polynomial seed and exact matching

We next construct the complex seed to which the local correction will be applied. Its roots must be sufficiently separated for independent replacements, while its inner and outer collars must match the exact models with negligible residual. We use degrees $k = q^2$ with $q \in \mathbb{N}$; the choice in Section 5 will show that these degrees suffice for every prescribed large radius. Set

$$\rho_j = 2 + \frac{j}{q+1} \quad (1 \le j \le q), \qquad P(z) = \prod_{j=1}^{q} (z^q - \rho_j^q). \tag{3.1}$$

Then P is monic of degree k, $P(0) \in \mathbb{R} \setminus \{0\}$, and its root set Λ consists of k simple points in $2 < |z| < 3$. For roots on different circles, the radial separation is at least $(q+1)^{-1}$; on one circle, the minimum distance is $2\rho_j \sin(\pi/q)$. Thus, for $q \ge 2$,

$$|\xi - \eta| \ge c_* k^{-1/2} \qquad (\xi \ne \eta,\ \xi, \eta \in \Lambda) \tag{3.2}$$

with a fixed $c_* > 0$.

Choose $\varepsilon_0 > 0$ so small that the closed disks of radius $3\varepsilon_0 k^{-1/2}$ centered at the roots are pairwise disjoint and lie in $7/4 < |z| < 13/4$. For each $\xi \in \Lambda$, put

$$H_\xi(z) = \frac{z}{P(z)/(z-\xi)}, \qquad \ell_\xi(z) = \frac{H_\xi'(z)}{H_\xi(z)}. \tag{3.3}$$

The factor H_ξ is holomorphic and nonvanishing on its disk; it isolates the pole at ξ from the contribution of the remaining roots.

3.1. Logarithmic derivatives

Lemma 3.1. *Uniformly in ξ and on $B_{3\varepsilon_0 k^{-1/2}}(\xi)$,*

$$|\ell_\xi| \le Ck, \tag{3.4}$$

$$|\ell_\xi'| \le Ck \log k, \tag{3.5}$$

$$|\partial_z^{j-1} \ell_\xi| \le C_j k^{j/2} \qquad (3 \le j \le 9). \tag{3.6}$$

Proof. Root separation first gives a uniform packing estimate: the disks of radius a fixed multiple of $k^{-1/2}$ centered at the roots are disjoint, so

$$\#(\Lambda \cap B_t(z)) \le C(1 + kt^2). \tag{3.7}$$

For z in the disk associated with ξ, all other roots are a distance at least $d_0 = ck^{-1/2}$ away, after decreasing ε_0 if needed. All roots and these disks lie in a fixed bounded set. Decomposing that set into rings $2^\nu d_0 \le |z - \eta| < 2^{\nu+1} d_0$ gives

$$\sum_{\eta \ne \xi} |z-\eta|^{-s} \le Ck \sum_{\nu=0}^{O(\log k)} (2^\nu d_0)^{2-s} \le \begin{cases} Ck, & s = 1, \\ Ck \log k, & s = 2, \\ C_s k^{s/2}, & s > 2. \end{cases}$$

The logarithmic loss is confined to the borderline case $s = 2$. Now use

$$\ell_\xi(z) = \frac{1}{z} - \sum_{\eta \ne \xi} \frac{1}{z - \eta}$$

and differentiate. Since z stays a fixed positive distance from zero, the root sums give all the required bounds; explicitly,

$$\partial_z^{j-1} \ell_\xi = (-1)^{j-1} (j-1)! \left(z^{-j} - \sum_{\eta \ne \xi} (z - \eta)^{-j} \right).$$

□

3.2. **Exponential matching to the exact models**

The same arrangement of roots also makes the seed exponentially close to its exact models on fixed collars. Define

$$E_-(z) = P(z)/P(0) - 1, \qquad E_+(z) = P(z)/z^k - 1,$$

with E_+ considered away from zero. We use the fixed enlarged collars

$$\Omega_- := \{9/8 < |z| < 13/8\}, \qquad \Omega_+ := \{27/8 < |z| < 31/8\}. \tag{3.8}$$

There are fixed $0 < \vartheta < 1$ and constants C_j such that

$$\begin{aligned} |D^j E_-| &\le C_j q \vartheta^q \quad \text{on } |z| \le 13/8, \\ |D^j E_+| &\le C_j q \vartheta^q \quad \text{on } |z| \ge 27/8, \end{aligned} \qquad 0 \le j \le 9. \tag{3.9}$$

Indeed,

$$\frac{P(z)}{P(0)} = \prod_{j=1}^{q} \left(1 - (z/\rho_j)^q\right), \qquad \frac{P(z)}{z^k} = \prod_{j=1}^{q} \left(1 - (\rho_j/z)^q\right). \tag{3.10}$$

On the larger regions $|z| \le 7/4$ and $|z| \ge 13/4$, respectively, every power has modulus at most ϑ_0^q for a fixed $\vartheta_0 < 1$. The product error is bounded by $\exp(q\vartheta_0^q) - 1 \le Cq\vartheta_0^q$ for large q. Cauchy estimates on disks of a fixed radius, contained in these larger regions, give (3.9). The same bounds hold for real partial derivatives because the errors are holomorphic.

Choose fixed smooth radial cutoffs taking values in $[0, 1]$ with

$$\begin{aligned} \eta_-(r) &= 1 \quad (r \le 5/4), & \eta_-(r) &= 0 \quad (r \ge 3/2), \\ \eta_+(r) &= 0 \quad (r \le 7/2), & \eta_+(r) &= 1 \quad (r \ge 15/4). \end{aligned}$$

Set

$$S = 1 + \eta_- E_- + \eta_+ E_+, \qquad F^0(z) = \frac{|z|^2}{P(z)} S(z) \quad (z \notin \Lambda). \tag{3.11}$$

The outer cutoff term is set to zero near the origin. By (3.9), we have $|S - 1| \le 1/2$ everywhere for large k, whereas $S = 1$ on every root disk. Thus the collar modifications leave the local pole structure unchanged and give the exact models

$$\begin{aligned} F^0(z) &= P(0)^{-1}|z|^2 \quad (|z| \le 5/4), \\ F^0(z) &= |z|^2 z^{-k} \qquad (|z| \ge 15/4). \end{aligned} \tag{3.12}$$

Away from the roots, the only biharmonic residual is confined to the two closed transition collars

$$\mathcal{A}_- := \{5/4 \le |z| \le 3/2\}, \qquad \mathcal{A}_+ := \{7/2 \le |z| \le 15/4\}.$$

Their distances from the boundaries of Ω_- and Ω_+, respectively, are at least $1/8$. This fixed margin will permit the local correction without changing either exact model.

3.3. **Relative bounds and phase on the collars**

On each enlarged collar, F^0 is nonzero and satisfies

$$|D^j F^0| \le C_j k^j |F^0| \qquad (0 \le j \le 9), \tag{3.13}$$

$$|D^j \Delta^2 F^0| \le C_j k^{j+2} q \vartheta^q |F^0| \qquad (0 \le j \le 5). \tag{3.14}$$

To verify the second estimate, write $H = z/P$ and $G = \bar{z}S$ in (2.1). Since the collars stay a fixed distance from all roots and zero, logarithmic differentiation gives $|H^{(j)}| \le C_j k^j |H|$ for the required orders. Moreover,

$$\bar{\partial}^2(\bar{z}S) = 2\bar{\partial}S + \bar{z}\bar{\partial}^2 S.$$

Every nonzero term, including those obtained by further differentiation, contains a derivative of the exponentially small function $S - 1$. Applying (2.1) and then at most five additional derivatives gives (3.14). At most nine derivatives fall on the collar error, while the initial product identity places at most

two derivatives on H. The relative cost is therefore k^{j+2}. Finally, $|G|$ is bounded above and below by fixed positive constants on the collars, so the estimate may be divided by $|F^0| = |H||G|$.

On Ω_-, the formula can be written as

$$F^0 = P(0)^{-1}|z|^2(1+T_-), \qquad T_- = (\eta_- - 1)\frac{E_-}{1+E_-}.$$

Since $P(0)^{-1}$ is real and $|T_-| \le Cq\vartheta^q$,

$$|\operatorname{Re} F^0| \ge \frac{1}{2}|F^0| \quad \text{on } \Omega_- \tag{3.15}$$

for large k. The real part is bounded away from zero relative to its amplitude, so no zero-set correction is needed on the inner collar. On Ω_+,

$$F^0 = |z|^2 z^{-k}(1+T_+), \qquad T_+ = (\eta_+ - 1)\frac{E_+}{1+E_+}, \qquad |D^j T_+| \le C_j q\vartheta^q \quad (j \le 9).$$

Therefore

$$|\nabla \arg F^0| \ge \frac{k}{|z|} - \left|\frac{\nabla T_+}{1+T_+}\right| \ge ck \quad \text{on } \Omega_+. \tag{3.16}$$

Thus the inner collar permits direct division, while the outer collar has the phase transversality needed for correction. Both collars in (3.8) are disjoint from the root disks, and all preceding constants are independent of the root replacements. It remains to remove the poles without losing this separation or exceeding the potential budget.

4. **Adaptive removal of the singularities**

The replacement must do more than remove a pole: it must keep the real-part zero set transverse wherever a residual is created. We choose the radius so that either the holomorphic phase or the local replacement phase dominates the other. This two-scale choice prevents phase cancellation and keeps the residual within the required budget.

Fix a smooth function $\chi_0 : [0,\infty) \to [0,1]$ such that $\chi_0 = 0$ on $[0,1]$ and $\chi_0 = 1$ on $[2,\infty)$. For $\xi \in \Lambda$, write $w = z - \xi$ and define

$$G_{\xi,\varepsilon}(z) = \chi_0(|w|/\varepsilon)\frac{\bar z}{w} + (1 - \chi_0(|w|/\varepsilon))\frac{\bar\xi}{\varepsilon^2}\bar w. \tag{4.1}$$

The first summand is defined to be zero near $w = 0$.

Lemma 4.1. *On $\Omega_{\xi,\varepsilon} := \{\varepsilon/2 < |z-\xi| < 3\varepsilon\}$, uniformly for $2 \le |\xi| \le 3$ and all sufficiently small $\varepsilon > 0$,*

$$c\varepsilon^{-1} \le |G_{\xi,\varepsilon}| \le C\varepsilon^{-1}, \tag{4.2}$$

$$|D^j G_{\xi,\varepsilon}| \le C_j \varepsilon^{-j}|G_{\xi,\varepsilon}| \qquad (0 \le j \le 9), \tag{4.3}$$

$$\nabla \arg G_{\xi,\varepsilon} = -\nabla \arg w + O(1). \tag{4.4}$$

The $O(1)$ is a real vector with a bound independent of ξ, ε. Whenever H_ξ is defined, the product $H_\xi G_{\xi,\varepsilon}$ is biharmonic on $|w| < \varepsilon$ and on $|w| > 2\varepsilon$.

Proof. Write $w = se^{i\theta}$ and abbreviate $\chi_0(s/\varepsilon)$ by χ_0. Then

$$G = \bar\xi e^{-i\theta}B(s) + \chi_0 e^{-2i\theta}, \qquad B(s) = \frac{\chi_0}{s} + (1-\chi_0)\frac{s}{\varepsilon^2}.$$

On $\varepsilon/2 < s < 3\varepsilon$, the positive quantity B lies between c/ε and C/ε. Hence

$$G = \bar\xi e^{-i\theta}B(s)(1+T), \qquad T = \frac{\chi_0 e^{-i\theta}}{\bar\xi B(s)}, \qquad |T| \le C\varepsilon, \quad |\nabla T| \le C. \tag{4.5}$$

To estimate the derivative in (4.5), write $B(s) = \varepsilon^{-1}b(s/\varepsilon)$. The positive function b and its reciprocal have uniformly bounded derivatives on $[1/2, 3]$. Thus T is ε times a smooth profile of w/ε, which gives

$|\nabla T| \le C$. For small ε, the factorization (4.5) yields (4.2). Since the radial factor B is real and positive, it contributes no phase:

$$\nabla \arg G = -\nabla\theta + \operatorname{Im} \frac{\nabla T}{1+T},$$

which proves (4.4).

For the higher derivative estimates, put $w = \varepsilon\zeta$ and write

$$G = \frac{\bar\xi}{\varepsilon}\left(\frac{\chi_0(|\zeta|)}{\zeta} + (1-\chi_0(|\zeta|))\bar\zeta\right) + \chi_0(|\zeta|)\frac{\bar\zeta}{\zeta}.$$

On the fixed ring $1/2 < |\zeta| < 3$, all derivatives through order nine of the two bracketed profiles are bounded. The chain rule gives $|D^j G| \le C_j \varepsilon^{-j-1}$, which, together with the lower bound in (4.2), yields (4.3). In the core and outside the transition ring, the exact profiles are $(\bar\xi/\varepsilon^2)\bar w$ and $\bar z/w$, respectively. Both are annihilated by $\bar\partial^2$ on their domains, so (2.1) proves exact biharmonicity in both regions. □

The radius is now chosen from the strength of the holomorphic phase. Define $a_\xi = |\ell_\xi(\xi)|$. With ε_0 fixed, choose a small constant $\varepsilon_1 > 0$ and set

$$\varepsilon_\xi = \begin{cases} \varepsilon_0 k^{-1/2}, & a_\xi \ge k^{2/3}, \\ \varepsilon_1 k^{-2/3}, & a_\xi < k^{2/3}. \end{cases} \tag{4.6}$$

For all sufficiently large k, every disk $B_{3\varepsilon_\xi}(\xi)$ is contained in its original separation disk. In these disks replace F^0 by $H_\xi G_{\xi,\varepsilon_\xi}$, and retain F^0 elsewhere. Since $S = 1$ on the root disks, the formulas agree on the open rings $2\varepsilon_\xi < |z-\xi| < 3\varepsilon_\xi$. They therefore define a global smooth complex-valued function F, with no residual introduced at the gluing boundaries.

Proposition 4.2. *For each enlarged root ring $\Omega_\xi := \{\varepsilon_\xi/2 < |z-\xi| < 3\varepsilon_\xi\}$, the function F is nonvanishing and there are $L_\xi \ge 1$, $D_\xi > 0$ such that*

$$|D^j F| \le C_j L_\xi^j |F| \qquad (0 \le j \le 9), \tag{4.7}$$

$$|D^j \Delta^2 F| \le C_j D_\xi L_\xi^j |F| \qquad (0 \le j \le 5), \tag{4.8}$$

$$|\nabla \arg F| \ge c L_\xi, \qquad D_\xi \le C L_\xi^4, \qquad D_\xi \le C k^3. \tag{4.9}$$

The constants are uniform in k and ξ. The residual is supported in $\{\varepsilon_\xi \le |z-\xi| \le 2\varepsilon_\xi\}$, at distance at least b/L_ξ from $\partial\Omega_\xi$, for a fixed $b > 0$.

Proof. We first separate the phase estimate from the derivative costs. For a nonvanishing holomorphic function H, with $\ell = H'/H$,

$$\nabla \arg H = (\operatorname{Im}\ell, \operatorname{Re}\ell), \qquad |\nabla \arg H| = |\ell|. \tag{4.10}$$

Moreover, $H^{(j)}/H$ is a polynomial in $\ell, \ell', \dots, \ell^{(j-1)}$, homogeneous of weight j when $\ell^{(r)}$ is assigned weight $r+1$. Consequently, the bounds $|\ell^{(r)}| \le C_r L^{r+1}$ for $0 \le r \le 8$ imply

$$|H^{(j)}| \le C_j L^j |H| \qquad (0 \le j \le 9). \tag{4.11}$$

Case A: $a_\xi \ge k^{2/3}$. Set $L_\xi = a_\xi$ and $\varepsilon_\xi = \varepsilon_0 k^{-1/2}$. Lemma 3.1 gives, throughout the corresponding disk,

$$|\ell_\xi(z) - \ell_\xi(\xi)| \le Ck \log k\, \varepsilon_\xi \le C\varepsilon_0 k^{1/2} \log k = o(k^{2/3}).$$

For all sufficiently large k, with a threshold independent of ξ, we therefore have

$$\frac{3}{4} L_\xi \le |\ell_\xi(z)| \le \frac{5}{4} L_\xi.$$

The higher logarithmic derivatives satisfy the weighted bounds needed for (4.11): indeed, $k \log k \le C L_\xi^2$ and $k^{j/2} \le L_\xi^j$ for $j \ge 3$. Moreover,

$$|\nabla \arg G_{\xi,\varepsilon_\xi}| \le \frac{2}{\varepsilon_\xi} + C = o(L_\xi).$$

The reverse triangle inequality and (4.10) therefore give

$$|\nabla \arg F| \ge |\ell_\xi| - |\nabla \arg G_{\xi,\varepsilon_\xi}| \ge \frac{1}{2} L_\xi. \tag{4.12}$$

Here the holomorphic phase dominates. No alignment of the two phase gradients is required; even opposite directions are allowed.

Case B: $a_\xi < k^{2/3}$. We now shrink the radius so that the replacement phase dominates. Set $\varepsilon_\xi = \varepsilon_1 k^{-2/3}$ and $L_\xi = \varepsilon_\xi^{-1}$. On the smaller disk,

$$|\ell_\xi(z)| \le k^{2/3} + C\varepsilon_1 k^{1/3} \log k \le 2k^{2/3}$$

for all sufficiently large k. The same weighted derivative bounds again imply (4.11) with $L = L_\xi$. On the enlarged ring, (4.4) gives

$$\begin{aligned} |\nabla \arg F| &\ge \frac{1}{3\varepsilon_\xi} - C - 2k^{2/3} \\ &= \left(\frac{1}{3} - 2\varepsilon_1\right) L_\xi - C. \end{aligned}$$

Choose $\varepsilon_1 < 1/24$, independently of k and ξ, and then increase k_0 so that $C \le L_\xi/8$. We obtain

$$|\nabla \arg F| \ge L_\xi/8. \tag{4.13}$$

Residual estimates. Since $\varepsilon_\xi^{-1} \le C L_\xi$, the relative derivative bounds (4.3) and (4.11) give (4.7). To estimate the residual, the three relative costs in (2.1) are

$$\frac{|\bar\partial^2 G|}{|G|} \le C\varepsilon_\xi^{-2}, \qquad \frac{|\partial\bar\partial^2 G|}{|G|} \le C\varepsilon_\xi^{-3}, \qquad \frac{|\partial^2\bar\partial^2 G|}{|G|} \le C\varepsilon_\xi^{-4}.$$

After $j \le 5$ additional real derivatives, the product rule yields

$$|D^j \Delta^2 (H_\xi G)| \le C_j L_\xi^j \big(L_\xi^2 \varepsilon_\xi^{-2} + L_\xi \varepsilon_\xi^{-3} + \varepsilon_\xi^{-4}\big) |H_\xi G|. \tag{4.14}$$

Each extra derivative costs at most $C L_\xi$; the largest derivative orders needed are seven for H_ξ and nine for G.

In Case A, $L_\xi \varepsilon_\xi \to \infty$, so the first term in (4.14) dominates. We take

$$D_\xi = L_\xi^2 \varepsilon_\xi^{-2}. \tag{4.15}$$

Since $L_\xi \le Ck$,

$$D_\xi \le Ck^3, \qquad D_\xi / L_\xi^4 \le Ck/L_\xi^2 \le Ck^{-1/3}.$$

In Case B all three terms have order L_ξ^4, so take

$$D_\xi = L_\xi^4 = \varepsilon_1^{-4} k^{8/3} \le Ck^3. \tag{4.16}$$

Since ε_1 is fixed, we in fact have $D_\xi/k^3 \to 0$ in this case. Thus the smaller radius still leaves room within the potential budget. Both choices give (4.8) and the remaining bounds in (4.9).

Finally, exact biharmonicity confines the residual to the stated closed transition ring. Its distance from $\partial\Omega_\xi$ is at least $\varepsilon_\xi/2$. In Case A, $L_\xi \varepsilon_\xi \ge \varepsilon_0 k^{1/6}$; in Case B, $L_\xi \varepsilon_\xi = 1$. A fixed $b > 0$ therefore gives the required margin for all sufficiently large k. □

4.1. **The real annular connection**

All regions carrying a residual now satisfy the hypotheses for real-part correction, except the inner collar, where the real part already stays away from zero. Because these regions are disjoint and the corrections preserve their boundary collars, the local potentials assemble without any loss proportional to the number of roots.

Proposition 4.3. *For every sufficiently large square integer $k = q^2$, there exist $f_k, Q_k \in C^\infty(\mathbb{R}^2;\mathbb{R})$ such that*

$$\Delta^2 f_k + Q_k f_k = 0, \qquad Q_k \in C_c^\infty(\{1 < |z| < 4\};\mathbb{R}), \qquad \|Q_k\|_\infty \le Ck^3, \tag{4.17}$$

and

$$\begin{aligned} &f_k(z) = A_k|z|^2 && (|z| \le 1),\\ &f_k(z) = \operatorname{Re}(|z|^2 z^{-k}) && (|z| \ge 4), \end{aligned} \qquad A_k = P(0)^{-1} \in \mathbb{R}\setminus\{0\}. \tag{4.18}$$

Proof. Start with $\operatorname{Re} F$, where F is the global smooth function from Section 4. On each Ω_ξ, apply Lemma 2.1 using Proposition 4.2. The estimates for $\Delta^2 F$ imply those for $E = \Delta^2 \operatorname{Re} F$, and $\operatorname{supp} E$ is contained in the same transition ring. All constants in the correction lemma are uniform in ξ, k.

On the outer enlarged collar Ω_+, apply the same lemma with

$$L = k, \qquad D = k^2 q\vartheta^q. \tag{4.19}$$

Equations (3.13)–(3.16) give its hypotheses. In particular, $D/L^4 = q\vartheta^q/k^2$ is bounded, and the residual support $\mathcal{A}_+$ has a fixed positive boundary margin. The lower bound on $\nabla \arg F^0$ holds on the whole enlarged collar.

The root disks are pairwise disjoint and avoid both Ω_- and Ω_+. Each correction is supported in the original residual support, compactly inside its enlarged region. Hence the corrected functions agree with $\operatorname{Re} F$ on open neighborhoods of every gluing boundary and define a single $f_k \in C^\infty(\mathbb{R}^2;\mathbb{R})$. Both exact models in (4.18) are preserved.

On Ω_-, no correction is made. By (3.15), we may define

$$Q_k = -\frac{\Delta^2 \operatorname{Re} F^0}{\operatorname{Re} F^0}, \qquad |Q_k| \le Ck^2 q\vartheta^q.$$

This is smooth and has support in $\mathcal{A}_-$. On every corrected region, use the smooth real potential supplied by Lemma 2.1. On the remaining open biharmonic regions, set $Q_k = 0$.

The potentials agree on overlaps because each vanishes in an open neighborhood of the boundary of its enlarged region. In particular, the core $|z-\xi| < \varepsilon_\xi$ retains the exact formula

$$f_k(z) = \operatorname{Re}\left(\frac{\bar\xi}{\varepsilon_\xi^2} H_\xi(z)(\bar z - \bar\xi)\right), \qquad \Delta^2 f_k = 0.$$

Setting $Q_k = 0$ in this open disk verifies the equation even at a critical zero of the real part; no division is performed there. The original replacements also agree on open overlap regions, so differentiation introduces no interface distributions.

The potential support is contained in

$$\mathcal{A}_- \cup \mathcal{A}_+ \cup \bigcup_{\xi\in\Lambda} \{\varepsilon_\xi \le |z-\xi| \le 2\varepsilon_\xi\}, \tag{4.20}$$

a compact subset of $1 < |z| < 4$. At each point at most one of these local potentials is nonzero. Hence

$$\|Q_k\|_\infty \le \max\left\{\sup_{\xi\in\Lambda} CD_\xi,\ Ck^2 q\vartheta^q\right\} \le Ck^3.$$

The estimate is independent of the number of roots. This proves (4.17). □

5. Kelvin inversion and normalization

The annular construction has supplied a real pair with an exact model at each end and a potential supported between them. Kelvin inversion exchanges these models, producing a high-order zero near the origin and a nonzero constant near infinity. Translation, normalization at a point of maximum modulus, and dilation will then place a unit observation ball at exactly the prescribed distance.

5.1. The transformed equation and its exact models

For $y \neq 0$, let $I(y) = y/|y|^2$. The planar biharmonic Kelvin identity is

$$\Delta_y^2\big(|y|^2 f(I(y))\big) = |y|^{-6}(\Delta^2 f)(I(y)). \tag{5.1}$$

For completeness, in polar coordinates put $D_r = r\partial_r$. Then

$$\Delta^2 = r^{-4}\big((D_r - 2)^2 + \partial_\theta^2\big)\big(D_r^2 + \partial_\theta^2\big).$$

With $s = r^{-1}$, the operator D_r acting on $r^2 f(s,\theta)$ corresponds to $2 - D_s$ acting on f. This substitution exchanges the two commuting factors. Hence

$$\Delta_y^2(r^2 f(s,\theta)) = r^{-2}(D_s^2 + \partial_\theta^2)((D_s - 2)^2 + \partial_\theta^2) f = r^{-2} s^4 \Delta_z^2 f,$$

which proves (5.1). The calculation is local away from the origin and requires only $f \in C^4$.

Apply it to Proposition 4.3, and define

$$g_k(y) = |y|^2 f_k(I(y)), \qquad \widehat{Q}_k(y) = |y|^{-8} Q_k(I(y)) \qquad (y \neq 0). \tag{5.2}$$

The exact models give

$$g_k(y) = \mathrm{Re}(\bar{y}^{\,k}) \quad (|y| \leq 1/4), \qquad g_k(y) = A_k \quad (|y| \geq 1). \tag{5.3}$$

Define $\widehat{Q}_k = 0$ at zero. Since it vanishes in a whole neighborhood of zero, both functions extend smoothly there, and

$$\Delta^2 g_k + \widehat{Q}_k g_k = 0, \qquad \operatorname{supp} \widehat{Q}_k \Subset \{1/4 < |y| < 1\}, \qquad \|\widehat{Q}_k\|_\infty \leq C_K k^3. \tag{5.4}$$

On the transformed support, $|y|^{-8} \leq 4^8$, so we may take $C_K = 4^8 C$, with C as in (4.17). The support restriction is essential: Kelvin inversion does not preserve boundedness for an arbitrary bounded potential. The exact polynomial near zero also rules out any point-supported distribution at the origin.

Let $M_k = \|g_k\|_\infty$. Evaluation at $y = (1/4, 0)$ yields $M_k \geq 4^{-k}$. There is a point y_k such that

$$1/4 \leq |y_k| \leq 1, \qquad |g_k(y_k)| = M_k. \tag{5.5}$$

Indeed, $|g_k(y)| \leq |y|^k < 4^{-k}$ for $|y| < 1/4$, while for $|y| \geq 1$ the function is constant and assumes that same value on $|y| = 1$. The supremum is therefore attained on the compact annulus in (5.5). This uses the exact models and compactness, not a maximum principle.

5.2. Dilation and completion of the proof

Proof of Theorem 1.1. We first work in dimension two. Fix $L_0 \geq \max\{1, C_K^{1/4}\}$. Given a sufficiently large real number R, choose the degree and dilation by

$$q = \big\lfloor (R/L_0)^{2/3} \big\rfloor, \qquad k = q^2, \qquad \lambda = R/|y_k|. \tag{5.6}$$

For $(R/L_0)^{2/3} \geq 2$,

$$\frac{1}{4L_0^{4/3}} R^{4/3} \leq k \leq \frac{1}{L_0^{4/3}} R^{4/3}. \tag{5.7}$$

Also $R \leq \lambda \leq 4R$ and

$$L_0 k^{3/4} = L_0 q^{3/2} \leq R \leq \lambda. \tag{5.8}$$

Define

$$u_R(x) = \frac{g_k(y_k + x/\lambda)}{g_k(y_k)}, \qquad V_R(x) = \lambda^{-4}\widehat{Q}_k(y_k + x/\lambda), \qquad x_R = -\lambda y_k. \tag{5.9}$$

The denominator is nonzero; dividing by its actual value fixes the sign at the origin. By (5.4) and (5.8),

$$\Delta^2 u_R + V_R u_R = 0, \qquad \|V_R\|_\infty \le C_K k^3 \lambda^{-4} \le 1.$$

The choice of y_k gives $u_R(0) = \|u_R\|_\infty = 1$, and $|x_R| = R$. The planar potential is smooth and compactly supported.

The chosen translation sends the observation ball into the polynomial region. More precisely, for $x = x_R + v$ with $|v| < 1$, the argument of g_k is v/λ; when $\lambda > 4$, (5.3) applies and gives

$$|u_R(x)| \le \frac{\lambda^{-k}}{M_k} \le (4/\lambda)^k \le (4/R)^k. \tag{5.10}$$

If $R \ge 16$, then $\log(R/4) \ge \frac{1}{2}\log R$. Together with (5.7), this proves (1.5), for example with $c = (8L_0^{4/3})^{-1}$.

For $n > 2$, set

$$u_R^{(n)}(x) = u_R^{(2)}(x_1, x_2), \qquad V_R^{(n)}(x) = V_R^{(2)}(x_1, x_2),$$

and append zero coordinates to x_R. Then $\Delta_n^2 u_R^{(n)} = \Delta_2^2 u_R^{(2)}$, and the global norms are unchanged. The projection of $B_1^{(n)}((x_R, 0))$ onto the first two coordinates is exactly $B_1^{(2)}(x_R)$. Continuity and independence of the remaining coordinates therefore give equality of the unit-ball supremum norms, proving the asserted smallness in every dimension. The higher-dimensional support is generally cylindrical, not compact.

For each fixed R, the planar solution is a nonzero constant outside a sufficiently large ball, by the exterior model in (5.3). This completes the proof. □

6. **Concluding remarks**

Remark 6.1 (Comparison with the second-order equation). The fourth-order mechanism is not merely the availability of a local jet correction. For holomorphic H, the identity

$$\Delta^2\big(H(z)(\bar{z} - \bar{\xi})\big) = 0$$

keeps the root cores exactly biharmonic. By contrast, the second-order identity is

$$\Delta\big(H(z)(\bar{z} - \bar{\xi})\big) = 4H'(z),$$

whose right-hand side need not vanish, even on the zero set of the real part. The same root-core construction therefore does not yield a bounded real potential for the second-order equation. The local identity $\Delta(h^2 a)|_{h=0} = 2a|\nabla h|^2$ does provide a second-order cancellation, but it does not make these cores harmonic. What matters here is that possible critical zeros lie in open regions where the biharmonic residual vanishes identically. This explains the present construction; it does not exclude other second-order methods.

Remark 6.2 (Finite-order zeros and the logarithm). Near zero, the transformed solution is exactly $g_k(y) = \mathrm{Re}(\bar{y}^k)$. Its leading jet has degree k, so its vanishing order is finite and exactly k; the construction produces neither an infinite-order zero nor a nontrivial compactly supported solution. For the final solution,

$$u_R(x_R + v) = \frac{\lambda^{-k}\,\mathrm{Re}(\bar{v}^k)}{g_k(y_k)} \qquad (|v| < \lambda/4)$$

in dimension two. The logarithm is thus a consequence of dilation at a zero of high finite order: $\lambda^{-k} = \exp(-k\log\lambda)$, with $k \asymp R^{4/3}$ and $\lambda \asymp R$. The normalization uses attainment of a maximum on a compact annulus, not a maximum principle for the biharmonic operator.

Remark 6.3 (Order of the parameter choices). The fixed radii, cutoffs and separation constant determine ε_0; then ε_1 is fixed and the uniform large-degree threshold is chosen. The resulting uniform constants in Lemma 2.1 determine δ, after taking the minimum of the root and outer-collar margin constants and the maximum of their derivative constants. The correction estimates determine C_K, which determines L_0. Only then is R_* fixed to ensure that (5.6) gives an admissible degree and that the final ball and logarithm estimates apply. In particular, neither δ nor the fixed dilation constant depends on k or R.

Remark 6.4 (Scalar classes and scope). The potentials in the final examples are scalar and real. Complex functions enter only as auxiliary devices in constructing that real pair. The inclusions

$$C_n^{\mathbb{R},\mathbb{R}}(C_0) \subset C_n^{\mathbb{C},\mathbb{R}}(C_0) \subset C_n^{\mathbb{C},\mathbb{C}}(C_0)$$

give the complex-class conclusion of Corollary 1.2 without a separate construction. The result concerns $\Delta^2 + V$ and does not extend, as stated, to arbitrary fourth-order elliptic principal parts, fixed-sign potentials, or classes with additional uniform regularity or support restrictions. Nor does it determine the optimal all-direction decay of a single fixed real solution.

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